

# HERMES Hardware Description

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February 20, 2026

## **Abstract**

This document describes the software setup of the sBitx HF transceiver. The sBitx radio is an ahead-of-its-time wide-band HF transceiver, fully controlled by software. The radio is composed by the analog radio-frequency circuitry, Raspberry Pi 4 (RPi 4), ATTiny85 microcontroller (for lambda bridge FWD and REF readings), battery-backed Real-Time-Clock, Wolfson wm8731 audio codec, 7 inches high-def touch screen and two knobs with “ button”,

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## 1 Introduction

This document address all the software details needed for the sBitx hardware bringup.

## 2 sBitx Hardware

### 2.1 Power consumption

## 3 Hardware interfaces to the computer (RPi 4)

### 3.1 Control I/O

#### 3.1.1 I2C RTC

#### 3.1.2 I2C Si5351

#### 3.1.3 I2C ATtiny85 (FWD and REF)

#### 3.1.4 Digital pins setup for knobs control

### 3.2 Signal I/O

#### 3.2.1 I2S Wolfson wm8371

## 4 Linux setup for sBitx operation

### 4.1 Pin attribution setup for the RPi 4

### 4.2 Audio - ALSA subsystem

### 4.3 Ashhar's sbitx reference implementation

#### 4.3.1 power setup

#### 4.4 bridge setup

#### 4.5 I2C bit-banging

#### 4.6 RTC setup